

COTULLA-LA SALLE COUNTY AP, TX, USA

WMO: 722526

Lat: 28.457N

Lon: 99.218W

Elev: 145

StdP: 99.59

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	2.0	-11.3	1.4	8.5	-8.9	1.8	11.2	10.8	17.4	9.0	16.3	2.0	50	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	39.7	22.9	38.7	23.0	37.7	23.1	26.1	31.7	25.7	31.4	25.3	31.0	4.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.3	27.3	24.4	19.7	27.1	24.0	19.2	27.0	81.4	31.9	79.6	31.5	78.1	31.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.4	-3.2	41.6	2.1	1.3	-4.7	42.5	-6.0	43.3	-7.1	44.0	-8.7	44.9		
			-4.8	27.1	2.0	0.8	-6.3	27.7	-7.4	28.2	-8.5	28.7	-10.0	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	13.1	15.7	19.4	23.4	26.9	29.8	30.2	30.8	28.0	23.7	18.1	13.8
	DBStd	7.39	5.08	5.52	4.70	3.90	3.17	1.82	1.87	1.65	2.61	4.16	4.99	4.89
	HDD10.0	64	26	13	3	0	0	0	0	0	0	4	18	
	HDD18.3	564	175	105	46	7	1	0	0	0	9	65	156	
	CDD10.0	4724	122	173	294	402	523	593	626	645	539	425	248	136
	CDD18.3	2183	13	32	78	159	265	343	367	387	289	176	58	15
	CDH23.3	25628	134	301	682	1563	2944	4398	4778	5293	3197	1738	471	129
	CDH26.7	13639	30	98	242	710	1511	2523	2777	3178	1669	743	133	23
Wind	WSAvg	3.6	3.0	3.4	3.8	4.1	4.3	4.2	4.0	3.7	3.1	3.2	3.0	2.8
Precipitation	PrecAvg	552	27	31	41	44	70	57	59	50	70	57	34	27
	PrecMax	955	111	89	110	125	194	162	271	188	195	234	111	110
	PrecMin	244	0	0	0	0	0	0	0	4	4	4	0	0
	PrecStd	171	27	30	33	38	53	48	70	57	50	54	32	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	32.8	33.9	37.9	39.1	40.9	40.6	40.9	39.4	35.8	32.2	29.2
		MCWB	17.7	17.5	18.7	21.2	23.2	22.9	22.9	23.0	22.0	22.8	21.1	18.9
	2%	DB	27.0	29.6	31.5	35.1	37.4	38.8	39.4	39.6	37.5	33.9	29.9	26.7
		MCWB	16.8	17.7	19.2	20.6	23.2	23.2	22.9	23.0	22.8	22.6	20.2	17.8
	5%	DB	24.3	27.2	29.4	32.9	35.8	37.6	38.2	38.5	36.0	32.4	28.0	24.5
		MCWB	15.7	17.6	18.9	21.1	23.0	23.2	23.1	23.2	22.9	21.9	19.7	17.2
	10%	DB	22.0	24.5	27.3	30.8	33.9	36.3	37.0	37.5	34.3	31.0	26.2	22.3
		MCWB	15.5	16.8	18.5	20.2	22.7	23.3	23.3	23.3	23.0	21.6	19.0	16.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	22.2	22.5	24.2	26.1	26.8	26.4	26.2	26.4	25.8	23.5	21.6
		MCDB	24.9	26.2	28.0	30.7	32.6	33.0	31.9	32.4	30.4	29.5	27.5	25.3
	2%	WB	19.6	20.7	21.7	23.5	25.4	25.9	25.8	25.7	25.6	24.9	22.5	20.6
		MCDB	23.5	25.3	26.7	29.8	31.8	32.1	31.5	31.8	30.1	29.0	26.5	24.2
	5%	WB	18.3	19.7	21.0	22.9	24.8	25.5	25.4	25.2	25.2	24.3	21.6	19.4
		MCDB	22.0	23.8	25.6	28.9	31.0	31.5	31.3	31.4	29.9	28.4	25.6	22.5
	10%	WB	16.7	18.7	20.3	22.3	24.2	25.0	25.1	25.0	24.7	23.7	20.6	17.9
		MCDB	20.4	22.5	24.8	27.8	30.2	30.9	30.9	31.2	29.5	27.8	24.6	21.2
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.8	12.7	13.2	12.7	13.0	12.7	13.3	11.7	12.5	12.4	12.5
		MCDBR	16.3	16.0	15.4	15.5	14.9	14.8	14.9	14.8	13.7	13.1	13.4	15.4
		MCWBR	8.0	7.2	6.0	5.0	3.8	3.0	3.0	3.0	3.3	4.2	5.8	7.8
	5% WB	MCDBR	11.9	12.1	11.3	12.2	12.0	12.4	12.2	12.9	10.9	10.7	11.0	11.3
	MCWBR	7.0	7.0	5.2	4.5	3.8	3.3	3.1	2.9	3.1	4.4	5.7	6.9	
Clear-Sky Solar Irradiance	taub	0.337	0.342	0.362	0.400	0.430	0.417	0.417	0.415	0.404	0.368	0.354	0.328	
	taud	2.474	2.466	2.424	2.336	2.300	2.392	2.400	2.413	2.396	2.489	2.464	2.515	
	Ebn at Noon	905	930	928	897	866	871	870	872	873	890	877	899	
	Edh at Noon	94	103	114	128	133	121	120	117	116	99	94	87	
All-Sky Solar Radiation	RadAvg	3.10	3.71	4.67	5.50	5.94	6.79	6.75	6.50	5.32	4.45	3.38	2.79	
	RadStd	0.38	0.66	0.54	0.56	0.60	0.39	0.54	0.32	0.47	0.57	0.40	0.42	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+242
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+96

Nomenclature: See separate page